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Embedded Declaratives in Korean*

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Jae-Il Yeom. 2018. Embedded Declaratives in Korean. *Language and Information, 22.1, 1-27.* This paper deals with three verbs that take a declarative clause. The verb *al* 'know' takes a *ko*-phrase, which is non-factive, and a *kes*-phrase, which describes a fact. The verb *mit* 'believe' also takes both, but a *kes*-phrase taken by the verb denotes a proposition. The verb *sayngkakha* 'think' takes a *ko*-phrase, but not a *kes*-phrase. I claim that a *ko*-phrase is a predicate which reflects what is in the mind of the subject. A *kes*-phrase expresses something that exists independently of the epistemic process. It can be a fact or a proposition that is given. If a *kes*-phrase is selected by *al*, it denotes a fact and if a *kes*-phrase is selected by *mit*, it denotes a proposition. *sayngkakha* only expresses an epistemic state, and it does not take a *kes*-phrase. I also discuss an adnominal declarative clause as part of a *kes*-phrase.

Key words: embedded clause, *ko*, *kes*, declarative, complement clause

1. Introduction

In this paper, I will consider three verbs that can take an embedded declarative clause that denotes a fact or proposition, In Korean, there are various forms of clausal structure that can denote a fact or proposition, but I focus on three forms of them. One structure that can be considered as an internal argument of the three verbs is a clause that ends with ko:[1]

(1) inho-nun mina-ka ttena-ss-ta-ko {al, mit, sayngkakha}-ko.iss-ta.
Inho-top Mina-nom leave-pst-dec-cmp {know believe think}-prog-dec

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1 In this paper I use the following glosses: acc(usative case), adn(ominal ending), adz = ad(nominali)z(er), cmp = c(o)mp(lementizer), dec(larative mood), hor(tative mood), imp(erative mood), int(errogative mood), nml = n(o)m(ina)l(izer), nom(inative case), npst = n(on-)p(a)st, pst = p(a)st, pl(ural marker), prog(ressive), res.st = result state, retro(spective), top(ic marker), etc.

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'Inho {knows, believes, thinks} that Mina left.'

Here *ko* combines with a Mood Phrase (= MP), which can be used as a matrix clause, and forms an embedded clause. A *ko*-phrase always describes a non-factual situation, and even the verb *al* 'know' takes a *ko*-phrase.

If a *ko*-phrase could be used really as an argument of a verb, it would be expected that a *ko*-phrase is also used as the external argument of a verb. However, a clause ending with *ko* cannot be used in the subject position. Instead, a *kes*-phrase is used:[2]

(2) mina-ka ttena-ss-ta-{*ko-ka, nun kes-i} nollawu-ess-ta.
Mina-nom leave-pst-dec-{cmp-nom, adn thing-nom} $_{\text{be}}$surprising[3]-pst-dec
'The thing that Mina left was surprising.

This casts doubt on the idea that a *ko*-phrase in the complement position of a verb is an internal argument of the verb.

A *kes*-phrase can denote various semantic entities including objects/individuals, events/situations, facts and propositions, but in this paper, I am only interested in *kes*-phrases that denote a fact or a proposition. See Asher (1993) for abstract semantic entities. In a *kes*-phrase, a clause can be embedded with or without a declarative Mood marker *ta*. The three verbs allow different forms of *kes*-phrases:[4]

(3) a. inho-nun mina-ka ttena-{n, ss-ta-nun} kes-ul al-ko.iss-ta.
Inho-top Mina-nom leave-{adn, pst-dec-adn} thing-acc know-prog-dec
'Inho knows that Mina left.'
b. inho-nun mina-ka ttena-{*n, ss-ta-nun} kes-ul mit-ko.iss-ta.

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2 In addition, if a *that*-clause is used in the subject position, it is often assumed that it is a DP. See Lees (1960), Rosenbaum (1967), Ross (1967), Hartman (2012), etc.

3 In Korean, there is no adjective that combines with a copula and forms a VP. Instead, Korean has a stative verb. If a Korean stative verb has a corresponding adjective A in English, I will gloss the verb as '$_{\text{be}}$A'.

4 One reviewer claims that (s)he does not agree with the acceptability here, but (s)he is not clear about which example (s)he has in mind. There is a possibility that (s)he interprets examples here in a different way that the way I do. For example, (3c) can be acceptable if the verb is interpreted as the one in the following example:

i. inho-nun {ecey il, emeni}-(l)ul sayngkakha-ko.iss-ta.
Inho-top {yesterday event, mother}-acc think-prog-dec
'Inho is thinking of {the event yesterday, his mother}.'

Here, *sayngkakha* 'think' has the meaning of 'think of', which I am not interested in in this paper.

Inho-top Mina-nom leave-{adn, pst-dec-adn} thing-acc believe-prog-dec
'Inho believes that Mina left.'
c. inho-nun mina-ka ttena-{*n, *ss-ta-nun} kes-ul sayngkakha-ko.iss-ta.
Inho-top Mina-nom leave-{adn, pst-dec-adn} thing-acc think-prog-dec
'Inho thinks that Mina left.'

The verb *al* takes two forms of a *kes*-phrase and denotes a fact, but a *kes*-phrase is used with mit only with the *ta-nun* form and it denotes a proposition. This needs to be explained. And *sayngkakha* 'think' is only used with a *ko*-phrase.[5] I suppose that the three verbs and the three clausal structures are subject to certain restrictions, and I will explain what the restrictions are.

The paper is organized in the following order. In Section 2, I discuss the interpretation of *ko*-phrases. I claim that a ko-phrase is in a predication relation with an overt or covert object. I suppose that this is related to the fact that to get an answer with a *ko*-phrase, we use a question word *ettehkey* 'how'. I also claim that if a verb takes a *ko*-phrase, the verb expresses an epistemic state due to an internal epistemic process. This is why *ko*-phrases are not factive. In Section 3, I discuss *kes*-phrases with and without the declarative Mood marker *ta*. A *kes*-phrase without *ta* denotes a property of situations but it can be converted to a fact. A *kes*-phrase with *ta* can denote a property of propositions or facts. I claim that the adnominalizer *(u)n* is an abstraction over semantic entities that can be denoted by the embedded declarative sentence. In so doing, I also claim that we do not need to distinguish abstract objects and their contents. In Section 4, I conclude the paper.

2. *Ta-ko* 'dec-cmp'

2.1. Basic properties of *ko*-phrases

Structurally, *ko* takes a Mood Phrase (= MP) headed by a Mood marker like *ta* 'declarative', *nya*'interrogative', *la* 'imperative', *ca* 'hortative', etc.:

(4) ... {ta, nya, la, ca}]MP-ko
{dec, int, imp, hor}-cmp

5 Here I only consider uses of *sayngkakha* meaning 'think (that)'. If the verb is interpreted as meaning 'think of', the *kes*-phrase without *ta* is acceptable. In this case, the actual complement of the verb is not realized.

A Mood marker determines a sentence type and a MP can be used as a matrix clause. If *ko* takes a MP, it has the function of conveying the content of a sentence.[6] But *ko* can even be used in conveying an actual or potential utterance directly.

Since *ko* has the function of conveying (the content of) a sentence, it is not guaranteed that the sentence is true. Therefore, it is natural that it is used with a non-factive verb like *mit* 'believe' or *sayngkakha* 'think'. Since a *ko*-phrase is also used with the verb *al* 'know', as shown in (1), *al* is not inherently factive. This shows that the verb *al* itself does not have the meaning of factuality. A use of a *ko*-phrase does not simply allow a non-factual interpretation but requires such a meaning:[7]

(5) ??mina-ka ttena-ss-ta. inho-nun mina-ka ttena-ss-ta-ko {al,
Mina-nom leave-pst-dec Inho-top Mina-nom leave-pst-dec-cmp know
mit, sayngkakha}-ko.iss-ta.
believe, think-prog-dec
'Mina left. Inho {knows, believes, thinks} that Mina left.'

If the content of the embedded clause is a fact, it is odd to use a *ko*-phrase.

This is contrasted with a case where *al* is used with a kes-phrase, as shown in (3.a), where the two forms of a *kes*-phrase lead to the meaning of factuality. If a *kes*-phrase does not denote a fact, it cannot be used with *al*:

(6) ??mina-nun ttena-ss-ta. kulentey inho-nun mina-ka camkkan oychwulha-{n,
Mina-top leave-pst-dec but Inho-top Mina-nom a.while go.out-{adn,
yess-ta-nun} kes-ul al-ko.iss-ta.
pst-dec-adn} thing-acc know-prog-dec
'Mina left. But Inho knows that Inho went out for a while.'

Here what Inho knows is not a fact, and the sentence is bad. This shows that a *kes*-phrase must denote a fact when used with al.

In addition to non-factuality, when *al* is used with a *ko*-phrase, the verb needs an aspectual morpheme *ko.iss*, as shown in (1). Without the morpheme, the sentence would include the

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6 Here the content of something is the proposition that it conveys.

7 One reviewer claims that this example is odd because the same content is repeated, but if the complement of a verb is factual, it is fine to repeat the same content:
i. mina-ka ttena-ss-ta. inho-nun mina-ka ttena-n kes-ul al-ko.iss-ta.
Mina-nom leave-pst-dec Inho-top Mina-nom leave-adn thing-acc know-prog-dec
'Mina left. Inho knows that Mina left.'

morpheme *(nu)n* and the sentence becomes odd:

(7) ??na-nun Inho-ka haynypokha-ta-ko a(l)-n-ta.
I-top Inho-nom $_{be}$happy-dec-cmp know-npst-dec
'I know that Inho is happy.'

The morpheme *ko.iss* can be called a progressive aspectual morpheme. It generally expresses that the subject is in the middle of a single event or in the middle of repeating events. But the interpretation of an aspectual morpheme depends on what aspectual properties a verb it combines with has. If *ko.iss* combines with *al*, it makes the verb stative, excluding the phase of coming to know. On the other hand, the morpheme *(nu)n* expresses a non-past event. It can be used with a non-stative verb when the clause expresses an imperfective situation, a future situation, a generic situation, etc. In (7), *(nu)n* can be said to have the meaning of imperfectiveness, but this also makes the verb stative. We need to know what difference the two combinations make. I will discuss this question.

The verb *al* used with a *ko*-phrase is contrasted with the other two verbs or with a case where *al* takes a *kes*-phrase:

(8) inho-nun mina-ka ttena-ss-ta-ko {mit, sayngkakha}-{(nu)n, ko.iss}-ta.
Inho-top Mina-nom leave-pst-dec-cmp {believe think}-{npst, prog}-dec
'Inho {believes, thinks} that Mina left.'
(9) inho-nun mina-ka ttena-{n, ss-ta-nun} kes-ul a(l)-{n, ko.iss}-ta.
Inho-top Mina-nom leave-{adn, pst-dec-adn} thing-acc know-{npst, prog}-dec
'Inho knows that Mina left.'

Mit 'believe' and *sayngkakha* 'think' can be used with *(nu)n* or *ko.iss* if they take a *ko*-phrase, as in (8), and so can *al*, as in (9), if *al* takes a *kes*-phrase.

Then the differences between the two uses of *al* can be summarized as follows:

1. When al_1 is used with a *ko*-phrase, the verb needs an aspectual morpheme *ko.iss*. On the other hand, al_2, *mit* and *sayngkakha* can occur with *(nu)n* as well as *ko.iss*.
2. al_1 selects for a *ko*-phrase, which denotes a non-factual content, and al_2 selects for a *kes*-phrase, which denotes a fact.

The question is how non-factuality is related to the use of a progressive aspect in the uses of *al*. I suppose that this is related to the meaning of *ko*. We saw in (5), in contrast with (6), that when a *ko*-phrase is used, it cannot denote a fact. The reason is that a *ko*-phrase conveys the content of an internal state or internal process of knowledge or belief.[8] On the other hand, one crucial difference between knowledge and belief is this. A belief can come into being in the agent's own mind with or without an external source.[9] Even if there is an external source, what an agent believes arises in the agent's mind as a result of the agent's (internal) epistemic process. On the other hand, knowledge must have an external source and what the agent knows exists independently of the agent's cognitive process:[10]

(10)

	object of an epistemic process	epistemic state
knowledge	outside	inside
a belief	inside	inside

Given that a *ko*-phrase denotes something internal, if *mit* is used with a *ko*-phrase without *ko.iss*, it means that with or without an external source, a belief arises in the agent's mind, and that is what the agent believes. When the verb is used with *ko.iss*, the process of getting a belief is not the speaker's concern. The sentence just describes the (internal) state of having the belief. In contrast, the peculiarity of *sayngkakha* is that it is only used as describing an epistemic state. On the other hand, if *al* is used without *ko.iss*, the verb expresses the fact that an agent has gotten the knowledge of an external object. But if used with *ko.iss*, the sentence expresses the agent's cognitive state of having knowledge of the external object, without being concerned with the process of getting the knowledge.

These differences are reflected in the differences of acceptability in combining with aspectual adverbials *imi* 'already' and *icey* 'now':

(11) a. na-nun icey kulehkey {??a(l), mit, ??sayngkakha}-n-ta.
I-top now so { see, believe, think}-npst-dec
Now I {see, believe, think} so/it.
b. ??na-nun icey kulehkey {a(l), mit, sayngkakha}-ko.iss-ta.
I-top now so {know, believe, think}-prog-dec
Now I {see, believe, think} so/it.

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8 One reviewer asks if there is independent evidence for this claim. I give evidence for this claim later in this paper.

9 Syntacticians normally consider the agent to have the θ-role of experiencer of the verb(s). But I will follow the tradition of semantics by calling it an agent.

10 Recently, Kratzer (2006, 2016) and Moulton (2009, 2015) claim that complement clauses of propositional attitude verbs do not denote propositions but objects whose contents are propositions. I suppose that this idea should not be generalized to cover all propositional attitude verbs.

(12) a. na-nun imi kulehkey {??a(l), mit, sayngkakha}-n-ta.
I-top already so { see, believe, think}-npst-dec
'I already {see, believe, think} so/it.'
b. na-nun imi kulehkey {a(l), mit, sayngkakha}-ko.iss-ta.
I-top already so {know, believe, think}-prog-dec
I already {know, believe, think} so/it.

Here *kulehkey* 'so' can be assumed to substitute for a *ko*-phrase. The use of *icey* 'now' indicates the resulting state of a process of getting knowledge or a belief and the use of *imi* 'already' indicates a state of having knowledge or a belief, ignoring the process of getting knowledge or a belief. These observations show that *al* inherently expresses an epistemic process and *sayngkakha* expresses an epistemic state. On the other hand, *mit* can express both an epistemic process and an epistemic state. And the use of *ko.iss* makes an epistemic process verb into an epistemic state predicate.

Next I will discuss the dual selectional properties of *al* and *mit*. I will start with *al*. There are two uses of *al* and the two uses lead to different structural and semantic properties. I have not discussed yet what a *ko*-phrase denotes, but the most plausible candidate at the moment is a proposition, though I will propose a different analysis below. And there are two uses of *al* in terms of selectional properties. One is a proposition-selecting verb and the other a fact-selecting verb. To make the semantics work, we need to define relations between possible worlds, facts and situations. A possible world consists of situations, and a possible world is a maximal situation in it. A true proposition in a possible world is a fact in that possible world. On the other hand, a fact is the existence of a situation that verifies the corresponding proposition. Thus we can assume that a possible world can be modeled as a set of facts:

(13) a. A possible world consists of situations, and a possible world is a maximal situation in it.
b. For a possible world w, the existence of a situation s in w is a fact.
c. For a proposition p, if p is true in w, then there is a fact f in w that corresponds to p. (I will represent it p_w.)

In this paper, a fact is defined more generally. A fact does not have to belong to the actual world. I assume that a possible world consists of facts, whether it is the actual world or not. If situation s_1 includes situation s_2, I assume that the existence of s_1 (= f_1) includes the existence of s_2 (= f_2): That is, if $s_1 \sqsubseteq s_2$, then $f_1 \sqsubseteq f_2$. With facts as a separate type of semantic entities, the two meanings of *al* can be defined as follows:

(14) a. $[\![\text{al}_1]\!]^w = \lambda p \lambda x \lambda e[\text{believe}^w(e, x, p)]$ (Temporary)
b. $[\![\text{al}_2]\!]^w = \lambda f \lambda x \lambda e[\text{knoww}(e, x, f)]$ (f (= p_w): a variable over facts)

The verb *al* can take a proposition or a fact.

One problem with the meaning specifications in (14) is that the meaning of al_1 is not distinguished from that of *mit* 'believe' or *sayngkakha* 'think'. Moreover, *mit* can take a *ko*-phrase or a *kes*-phrase. The two uses of *mit* can be distinguished as mit_1, which takes a *ko*-phrase, and mit_2, which takes a *kes*-phrase. When mit_2 takes a *kes*-phrase, the *kes*-phrase is taken to denote a proposition. Therefore the meaning of a *ko*-phrase must be defined in a different way. This is what I am going to discuss in the next subsection. In the meaning of al_2, a fact exists independently of the event/situation of knowing and the verb expresses a relation between the subject and the fact. Similarly, mit_2 takes a *kes*-phrase, which denotes a proposition that exists independently of the event of believing. This difference will be discussed below.

2.2. Predicate analysis of *ko*-phrases

In the previous subsection, I claimed that if a *ko*-phrase is used with *al*1, it introduces a proposition which holds in an agent's knowledge state. But it is not clear what function it plays in a sentence. One way to see is to consider a question-answer pair. If a phrase were an object of a verb, it would be possible to ask a question with mwues 'what'. But (15) shows that a *ko*-phrase is not an object:

(15) A: inho-nun mwues-ul {al, mit, (?)sayngkakha}-ko.iss-nya?[11]
Inho-top what-acc {know, believe, think}-prog-int
'What does Inho {know, believe, think}?'
B: ??mina-ka ttena-ss-ta-ko {al, mit, sayngkakha}-ko.iss-ta.
Mina-nom leave-pst-dec-cmp {know, believe, think}-prog-dec
'He {knows, believes} that Mina left.'

When A asks with *mwues* 'what', B cannot answer with a *ko*-phrase. When we want to get an answer with a *ko*-phrase, we use an adverbial expression *ettehkey* 'how':

11 When *sayngkakha* is used with *mwues*, the verb has a non-stative meaning of using one's mind actively. It does not express a belief state.

(16) A: inho-nun ettehkey {al, mit, sayngkakha}-ko.iss-nya?
Inho-top how {know, believe, think}-prog-int
'How does Inho {know, believe, think}?' (Lit.)
'What does Inho think?' (Int.)
B: (inho-nun) mina-ka ttena-ss-ta-ko {al, mit, sayngkakha}-ko.iss-ta.
(Inho-top) Mina-nom leave-pst-dec-cmp {know, believe, think}-prog-dec
'He thinks that Mina came.'

This shows that a *ko*-phrase cannot be used as an object of a verb. We also saw in (2) that a *ko*-phease cannot be used as the subject of a sentence. This shows that a *ko*-phrase is not an argument.

When the expression *ettehkey* 'how' is used as an adverbial, it might ask a manner, but the use of the expression we consider here is not a question word for a manner:

(17) A: inho-nun ettehkey {al-ko.iss, ??a(l)}-nya?
Inho-top how {know-prog, know}-int
'How does Inho know?' (Lit.)
'What does Inho think?' (Int.)
B: mina-ka o-ass-ta-ko al-ko.iss-ta.
Mina-nom come-pst-dec-cmp know-prog-dec
'He thinks that Mina came.'

In (17), the intended meaning of A's statement is to describes what knowledge state Inho is in, with no implication of changing from one knowledge state to another. That meaning is expressed with *ko.iss*. Since no event is involved, *ettehkey* cannot be understood as a manner adverbial. Without *ko.iss*, *ettehkey* 'how' is interpreted as asking about the way that Inho gets the information about something.

To explain what function a *ko*-phrase does, it is helpful to see that as an answer to A's question in (16), B's statement can be expressed in the following way:

(18) inho-nun mina-ka ttena-n kes-ulo {al, mit, sayngkakha}-ko.iss-ta.
Inho-top Mina-nom leave-adn thing-as {know, believe, think}-prog-dec
'He thinks that Mina came.'

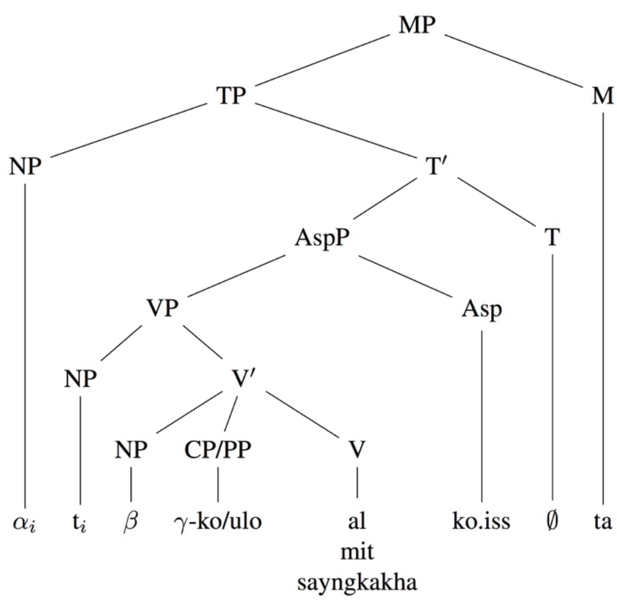
The expression *ulo* 'as' is a P that takes an NP. The question is what semantic role the *ulo*-phrase has. To answer this question, we need to look at some related constructions. The verbs in (16) can be used in the following construction:[12]

(19) A: ne-nun inho-lul ettehkey {al, mit, sayngkakha}-ko.iss-nya?
you-top Inho-acc how {know, believe, think}-prog-int
'What do you {know, believe, think} of Inho?'
B: na-nun inho-lul solcikha-n salam-{i-la-ko, ulo} {al, mit,
I-top Inho-acc behonest-adn person-{be-dec-cmp, as} {know, believe,
sayngkakha}-ko.iss-ta.
think}-prog-dec
'I think of Inho as an honest person.'

In this construction, we can suppose that there is a semantic relation of predication between *inho-lul* and *solcikha-n salam-{i-la-ko, ulo}* 'as an honest person'. Therefore it is plausible that the *ulo*-phrase and the *ko*-phrase are predicates.[13]

Since the syntactic structure is not my concern, I assume without further discussions that the object and the predicational phrase do not form a constituent:

(20)



12 One reviewer says that *mwues-ulo* 'as what' can replace *ettehkey* 'how', but the latter is more colloquial.
13 There are previous claims that *ko* is an adverbial ending. See Nam, K. and Ko, Y. (1985, 1998: 99-101) and Pak, M. (1993), etc., for such analyses.

In this structure *ko* can be considered either a complementizer or a P. One possible piece of evidence is that *ko* can introduce an adjunct clause that expresses a reason.[14] However, I will not take a definite position on the syntactic category of *ko* here.

The predication relation can hold between clauses too. We suppose that B's statement in (16) and the sentences in (18) are derived from something like this:

(21) inho-nun mina-ka ttena-n kes-ul mina-ka camkkan oychwulha-{n
Inho-top Mina-nom leave-adn thing-acc Mina-nom a.while go.out-{adn
kes-ulo, yess-ta-ko} {al, mit, sayngkakha}-ko.iss-ta.
thing-as, pst-dec-cmp} {know, believe, think}-prog-dec
'He thinks of Mina's leaving that she went out for a while.'

The fact that the verbs can take an object in addition to a *ko*-phrase or *ulo*-phrase indicates that the *kes/ulo*-phrase is not the object of the verbs. And if no overt object occurs, as in (18), we can assume that there is a null object in the sentence. Actually, it is common not to contain an overt object. It is cumbersome to include two embedded clauses in a row, after all.

I assume that the (syntactic) object denotes a fact. A fact can be understood as the existence of a situation, or as the corresponding proposition that is true in the actual world. Since a fact is linked to one possible world, it cannot be directly related to the doxastic alternatives of the syntactic subject. We need to consider the existence of situations in doxastic alternatives that correspond to the existence of the situation in the actual world. This cannot be done simply by a proposition that corresponds to the overtly expressed fact, because in the doxastic alternatives the proposition corresponding to the fact does not hold. Instead, we can think of a minimal fact that links the overtly expressed fact and the proposition that the subject believes is true. In (21), for example, the fact in the actual world that Mina left cannot be regarded as corresponding to the facts in doxastic alternatives that she went out for a while.

14 Here is one example:

i. inho-nun mina-ka swukoha-yess-ta-ko kunye-eykey yongton-ul cwu-ess-ta.
Inho-top Mina-nom make.efforts-pst-dec-cmp she-to allowance-acc give-pst-dec
'Inho gave Mina some pocket money because she made a lot of effects.'

I glossed *ko* as a complementizer but it has the meaning of 'because'. Therefore it is plausible to assume that it is a P. One common aspect of this use of *ko* and the one we are concerned with in this paper is that the use of the combination *ta-ko* conveys the meaning that there is a speech act involved. In Washo, an Amerindian language spoken in northern California and Nevada, USA, a non-factive embedded declarative clause ends with the dependent mood marker *-a*, and the same morpheme is otherwise used only in adjuncts. See Jacobsen (1964).

But we can think of a minimal fact that Mina did not show up for a while, with it being indeterminate whether she left or went out for a while. The minimal fact connects the two larger facts. The *ko*-phrase specifies the property of the minimal fact in terms of a part-of relation '$\sqsubseteq$':[15]

(22) $[\![[\alpha]\text{-(l)ul } [\beta]\text{-ko al1/mit1/sayngkakha}]\!]^w$
$= \lambda x\{[\![\alpha]\!]^w\}\exists p\forall w' \in \text{Dox}(w, x)[p_{w'} \sqsubseteq [\![\alpha]\!]_{w'} \wedge p_{w'} \sqsubseteq [\![\beta]\!]_{w'}]$
$\text{Dox}(w, x) = \{w \in W : w \text{ is compatible with what } x \text{ believes in } w\}$

Here p is a proposition corresponding to a minimal fact that allows us to capture the correspondence relation between the actual fact and the corresponding "facts" in the subject's doxastic alternatives. We do not consider epistemic alternatives because β is not really considered a fact in the actual world. In all doxastic alternatives, a fact p_w' is part of the "fact" $[\![\beta]\!]_w'$ in w'. This has the same effect that β entails p and that both of them hold in every doxastic alternatives.[16] There are cases where α can be a null object whose value is given in the context.

One remaining problem is what is the difference between *ko al* and *ko mit/ko sayngkakha*. We cannot distinguish epistemic alternatives from doxastic alternatives because in this use *al* is not factive. The former is used when there is one or more external sources of the knowledge and the sources provide the same content of knowledge. The agent is not sure that what she knows is true because there is a possibility the truth can be different from what she knows. The latter is used when the sources of the belief can be external or internal to the agent who has the belief and if they are external, they provide inconsistent contents. I will not formalize these pragmatic differences as part of their meanings.

3. Three *kes*-phrases

3.1. *Kes*-phrases without *ta*

We saw that *al* and *mit* can combine with a *kes*-phrase, while *sayngkakha* cannot. This is

15 Here the meaning of β is supposed to be a property of p_w' and a property is defined as a membership relation. But in the meaning specification, a part-of relation is used instead.

16 I will not discuss the meaning of a *ulo*-phrase because it is not my concern in this paper. I can suggest that it can be interpreted in a similar way to the one we interpret a *ko*-phrase. But the restriction on the use of *ulo*-phrases is not the same as that of *ko*-phrases. One piece of evidence is that *al* can be used without *ko.iss*.

related to the following:

(23) inho-nun mina-uy cwucang-ul {a(l), mit, *?sayngkakha}-(nu)n-ta.
Inho-top Mina-of claim-acc {know, believe, think}-npst-dec
'Inho {knows, believes, thinks} Mina's claim.' (Lit.)

Here Mina's claim exists independently of the situation denoted by the verbs. In this case, *sayngkakha* cannot be used. This means that *sayngkakha* only combines with a phrase that denotes something that comes into being in a person's internal epistemic process. This implies that a *kes*-phrase is expected to denote something that exists independently of a person's epistemic process. This results in the following differences:

(24) na-nun kukes-ul {a(l), mit, ??sayngkakha}-n-ta.
I-top it-acc {know, believe, ??think}-npst-dec
'I {know, believe, ??think} it.'

This shows that *sayngkakha* only expresses a thought that arises internally to the agent's belief, and the thought does not exist independently of the agent's epistemic process. In this section I will discuss how a *kes*-phrase is interpreted.

In Korean, there are three *kes*-phrases that denote a fact or proposition:

(25) a. inho-nun mina-ka ttena-n kes-ul al-n-ta.
Inho-top Mina-nom leave-adn thing-acc know-npst-dec
'Inho knows that Mina left.'
b. inho-nun mina-ka ttena-ss-ta-nun kes-ul al-n-ta.
Inho-top Mina-nom leave-pst-dec-adn thing-acc know-npst-dec
'Inho knows that Mina left.'
(26) inho-nun mina-ka ttena-{*n, ss-ta-nun} kes-ul mit-nun-ta.
Inho-top Mina-nom leave-{adn, pst-dec-adn} thing-acc believe-npst-dec
'Inho believes that Mina left.'

In (25.a), the *kes*-phrase does not include the Mood marker *ta*. In (25.b), the adnominal clause includes the declarative Mood marker. In these two examples, the *kes*-phrases denote facts. On the other hand, in (26), the verb *mit* requires a proposition-denoting phrase. In this case, the adnominal clause needs a Mood marker *ta*. From these observations, we can say the following:

1. Without the declarative Mood marker, a *kes*-phrase denotes a fact.
2. With the declarative Mood marker, a *kes*-phrase denotes a fact or a proposition.

From the observations, a few questions arise. What are structural differences between the two *kes*-phrases and how does the *kes*-phrase with ta denote a fact and proposition? I will start with the structure of the *kes*-phrase without *ta*.

To explain how a *kes*-phrase can denote a fact, I need to explain what is the internal structure of an adnominal clause and how it is interpreted, though it could be tedious. I have discussed examples without distinguishing two adnominal endings so far. One is *nun*, which is attached to eventive verbs and expresses non-perfectiveness, and the other is *(u)n*, which is attached to stative verbs with no meaning change, or is attached to eventive verbs to express the resulting states. Since they are used distinctively depending on the aspectual properties of verbs, it is more likely that the difference *n* between the two adnominal endings is an aspectual morpheme and the rest *(u)n* is an adnominalizer (= adz). The actual form of the adnominalizer changes depending on whether the preceding morpheme ends with a vowel:

(27) mek-n-un ki-n-un
eat-asp-adz creep-asp-adz
'eating' 'creeping'

(28) a. stative
hayngpokha-∅-n
$_{\text{be}}$happy-∅-adz
'being happy'

b. eventive
mek-∅-un ki-∅-n
eat-res.st-adz creep-res.st-adz
'having eaten' 'having crept'

However, we need to assume other morphemes than an Asp(ect) in an adnominal clause because these adnominal endings are contrasted with a modal adnominal *(u)l*, which is used to express the futurity of an event/state.[17] In this respect, the adnominal morpheme is expected to include the morpheme of a Tense. In an adnominal clause, the past Tense of a verb can be expressed with the form of "V-*ess-te-n*" 'V-pst-retro-adz'. This indicates that an adnominal clause can have syntactic positions for the Tense and other categories too, and it is reasonable to assume that an adnominal clause also has a full structure except a Mood marker, assuming that the adnominalizer is a C:[18]

17 One thing to mention is that in Korean, a Tense morpheme is interpreted relatively to the Tense of the embedding clause.

18 An adnominal clause with an overt Mood marker is discussed in the next subsection.

(29) $[[[[\ \alpha\text{-}\{\varnothing,\ n\}]_{AspP}\ \text{-}\varnothing_{T}]_{TP}\ \text{-(u)n}]_{CP}\ \text{kes}]_{NP}$

Precise interpretation of this structure needs a long discussion, but to save the space, I will make it short and show that an adnominal clause without a declarative Mood marker denotes a property of situations:

(30) a. $[\![[_{VP}\ \alpha]]\!]^{w} = \lambda e[\alpha'^{w}(e)]$

b. $[\![[_{AspP}\ [_{VP}\ \alpha\ ]\ \beta]]\!]^{w}$

$= \lambda P \lambda t \lambda s \exists e[s \sqsubset w\ \wedge\ P(e)\ \wedge\ R(\tau(e),t)\ \wedge\ \sigma\bullet\tau(e)\ \sqsubseteq\ \sigma\bullet\tau(s)]([\![\alpha]\!]^{w})$

$= \lambda t \lambda s \exists e[s \sqsubset w\ \wedge\ \alpha'^{w}\ (e)\ \wedge\ R(\tau(e),t)\ \wedge\ \sigma\bullet\tau\ (e)\ \sqsubseteq\ \sigma\bullet\tau\ (s)]\ (R \in \{\sqsubset,\ \sqsupset\})$

c. $[\![[_{TP}\ \alpha\ [_{T}\ \beta_{i}]]]\!]^{wg} = \lambda Q \lambda s[Q(g(i))(s)](\alpha'^{w}) = \lambda s[\alpha'^{w}(g(i))(s)]$

d. $[\![[_{CP}\ \alpha\ [_{C}\ \text{(u)n}\ ]]]\!]^{w} = [\![\alpha]\!]^{w}$

e. $[\![[_{NP}\ \alpha\ [_{N}\ \text{kes}]]]\!]^{w} = [\![\alpha]\!]^{w}$

The standard semantics assumes that a VP denotes a property of "events", without distinguishing events and states. In addition, I assume a situation variable to capture the meaning of the following case:

(31) na-nun mina-ka thonghaynglyo-ul an nay-nun kes-ul po-ass-ta.
I-top Mina-nom toll-acc not pay-adn thing-acc see-pst-dec
'I saw Mina not pay the toll.'

No event described by the VP occurs and such a situation is witnessed by the speaker. Therefore we need a situation variable to capture a situation described by a clause. I will call what a VP denotes an event and what a clause denotes a situation, but the two belong to the same type of semantic entities. I distinguish them just for convenience. I capture the relation between an event described by a VP and a situation described by a clause in terms of their spatio-temporal relation ($\sigma\bullet\tau$). R is a relation between an event time and a reference time, which changes with the aspectual property of the VP. If the VP is stative, the event time includes the reference; otherwise, the opposite holds. Tense provides the reference time and I assume that its value is provided by an assignment function. Then we get a property of situations. I assume that the adnominalizer and *kes* are semantically null. Thus a *kes*-phrase without a declarative Mood marker denotes a property of situations.

On the other hand, a *kes*-phrase can denote a fact:

(32) ??mina-nun ttena-ss-ta. kulentey inho-nun mina-ka camkkan oychwulha-n
Mina-top leave-pst-dec but Inho-top Mina-nom a.while go.out-adn
kes-ul a(l)-n-ta.
thing-acc know-npst-dec
'Mina left. But Inho knows that Mina went out for a while.'

Since the *kes*-phrase describes what is not true in the actual world, the sentence is odd. A *kes*-phrase inherently denotes a property of situations, but when a *kes*-phrase is used as an argument, a property undergoes one of the following conversions, unless it is bound by an unselective binder:

(33) a. $P \Rightarrow \iota x[P(x)]$ (ι-operation)
b. $P \Rightarrow \lambda Q \exists x[P(x) \wedge Q(x)]$ ($\exists$-operation)
c. $P \Rightarrow \{\exists x[P(x)]\}$ ({ }: presupposition)

If a property of situations undergoes the ι-operation, we get a definite situation. If it undergoes the $\exists$-operation, we get an existential quantifier. One more operation for the cases we are concerned with is that the situation variable is existentially bound and the content is presupposed. This is the case where a *kes*-phrase denotes a fact. This should be distinguished from the ι-operation on a situation variable:

(34) inho-nun mina-ka hwumchi-n cek-i iss-nun kes-ul a(l)-n-ta.
Inho-top Mina-nom steal-adn time-nom exist-adn thing-acc know-npst-dec
'Inho knows that Mina has once/several times stolen something.'

In this case, we cannot think of a unique situation where Mina has stolen. It is more plausible to assume that there is one or more times when Mina stole something. Therefore we need to assume that the situation variable is bound by an existential quantifier and the content is presupposed. Thus (33.c) is independently motivated.

Considering the discussions so far, we can derive the meaning of (25.a):

(35) a. $[\![\text{mina-ka ttena-n kes}]\!]^{w} = \lambda s[\text{left}^{w}(s,\text{mina})] \Rightarrow \{\exists s[\text{left}^{w}(s,\text{mina})]\}$
b. $[\![(25.a)]\!]^{w} = \exists s[\text{know}^{w}(s,\text{inho},\{\exists s'[\text{left}^{w}(s',\text{mina})]\})$
$(= \exists s'[\text{left}^{w}(s',\text{mina})] \wedge f = \exists s'[\text{left}^{w}(s',\text{mina})] \wedge \exists s[\text{know}^{w}(s,\text{inho},f)])$

The *kes*-phrase is interpreted as a property of situations but it is converted into a presupposition of the existence of one such situation. This presupposition is incorporated as the object of the verb *al*. The same meaning can be paraphrased as the meaning in which the presupposition is projected, as in the parentheses in (35b).

3.2. *Kes*-phrases with *ta*

Next, I will consider the *kes*-phrases in (25.b) and (26), which include a declarative Mood marker *ta*. Its meaning can be taken to be '(asserting that) there is a situation that verifies the sentence in the actual world'. To reflect this meaning, we can assume that the meaning of *ta* is to close the situation variable by an existential quantifier:

(36) $[\![[_{TP}\ \alpha]\text{-ta}]\!]^{w} = \exists s[[\![\alpha]\!]^{w}(s)]$, where $[\![\alpha]\!]^{w}$ is a property of situations.

Then the embedded clause in (25.b) can be interpreted as follows:

(37) $[\![\text{mina-ka ttena-}[_{T}\ ss_{i}]]\!]^{wg} = \lambda s[leave^{w}(s,mina) \wedge \tau(s) \sqsubset g(i) < t_{u}]$
(τ(s): the time Mina left; t_u: utterance time)
$[\![\text{mina-ka ttena-ss-ta}]\!]^{wg} = \exists s[leave^{w}(s,mina) \wedge \tau(s) \sqsubset g(i) < t_{u}]$

The meaning of a declarative sentence is the truth-conditions of the sentence, and the truth-value is determined in a possible world. It can be treated as a fact in this paper.

Then what does an adnominal clause denote? Assuming that *kes* is semantically null, if we look at the meanings of *kes*-phrases, we can guess how an adnominal clause should be interpreted. First, a *kes*-phrase can denote a fact. This can be explained by the claim that the content of a *kes*-phrase is presupposed:

(38) inho-nun hayngpokha-ta. mina-nun inho-ka hayngpokha-ta-nun kes-ul
Inho-top $_{be}$happy-dec Mina-top Inho-nom $_{be}$happy-dec-adn thing-acc
al-ko.iss-ta.
know-prog-dec
'Inho is happy. Mina knows that Inho is happy.'

(39) ??yuna-nun inho-ka hayngpokha-ta-ko malha-yess-ta. na-nun ku
Yuna-top Inho-nom $_{be}$happy-dec-cmp say-pst-dec I-top the
mal-ul mit-ci anh-ciman, mina-nun inho-ka
statement-acc believe-nml not.do-but Mina-top Inho-nom
hayngpokha-ta-nun kes-ul al-ko.iss-ta.

${}_{be}$happy-dec-adn thing-acc know-prog-dec
'Yuna said that Inho is happy. I do not believe the statement, but Mina knows that Inho is happy.'

When it is known that Inho is happy, it is fine to use a *kes*-phrase. Moreover, it needs to be a fact that Inho is happy, as shown in (39). From these observations, we can conclude a *kes*-phrase presupposes the content described by the adnominal clause.

However, we also know from (26) that a *kes*-phrase with ta may not denote a fact:

(40) ??mina-nun yeyppu-ta. inho-nun mina-ka yeyppu-ta-nun kes-ul
Mina-top ${}_{be}$pretty-dec Inho-top Mina-nom ${}_{be}$pretty-dec-adn thing-acc
mit-nun-ta.
believe-npst-dec
'Mina is pretty. Inho believes that Mina is pretty.'

But if the content of the *kes*-phrase is mentioned as an embedded clause, the discourse becomes fine:

(41) yuna-nun inho-ka hayngpokha-ta-ko malha-yess-ta. mina-nun inho-ka
Yuna-top Inho-nom ${}_{be}$happy-dec-cmp say-pst-dec Mina-top Inho-nom
hayngpokha-ta-nun kes-ul mit-ess-ta.
${}_{be}$happy-dec-adn thing-acc believe-pst-dec
'Yuna said Inho was happy. Mina believed Inho was happy.'

That is, when a *kes*-phrase does not denote a fact but a proposition, the content of the *kes*-phrase needs to be mentioned in the previous context. But the *kes*-phrase is not allowed simply by the embedded clause in a previous sentence:

(42) ??yuna-nun inho-ka hayngpokha-ta-ko sayngkakha-yess-ta. mina-nun
Yuna-top Inho-nom ${}_{be}$happy-dec-cmp think-pst-dec Mina-top
inho-ka hayngpokha-ta-nun kes-ul mit-ess-ta.
Inho-nom ${}_{be}$happy-dec-adn thing-acc believe-pst-dec
'Yuna thought that Inho was happy. Mina believed that Inho was happy.'

Here the content of the *kes*-phrase is mentioned in an embedded clause of the first sentence, but the discourse is odd. The important thing is whether the *kes*-phrase denotes a statement.

(41) is acceptable becausethe *kes*-phrase presupposes something like 'it is said that Inho is happy.' This is what Bogal-Allbritten and Moulton (2016) observe: when a *kes*-phrase denotes a proposition, the subject of the embedding verb *mit* 'believe' knows that there is an event of assertion.

From these observations, we can say the following about *kes*-phrases:

(43) a. A *kes*-phrase denotes something that exists independently of the meaning of the verb that takes it.
b. If a *kes*-phrase denotes a fact, the content is true in the actual world.
c. If a *kes*-phrase denotes a proposition, there is an event of assertion in the actual world.

These explain why *al* and *mit* can take a *kes*-phrase but *sayngkakha* cannot. The former have an external source, while the latter expresses an (internal) epistemic state.

The use of a *kes*-phrase presupposes its content or the assertion of its content. The question is, when a *kes*-phrase denotes a proposition, why does it require that there be an event of assertion? One tentative idea is that *ta-nun* abstracts over statements because *ta-nun* has come from *ta-ko ha-nun* 'dec-cmp do-adn', where the verb *ha* corresponds to a verb like *malha* 'say' or *cwucangha* 'assert/claim'.[19] It is really the case that (41) can be expressed with no meaning change as follows:

(44) yuna-nun inho-ka hayngpokha-ta-ko ha-yess-ta. mina-nun inho-ka
Yuna-top Inho-nom $_{\text{be}}$happy-dec-cmp do-pst-dec Mina-top Inho-nom
haynqpokha-ta-(ko ha)-nun {kes, mal, cwucang}-ul mit-ess-ta.
$_{\text{be}}$happy-dec-(cmp do)-adn {thing, statement, assertion}-acc believe-pst-dec
'Yuna said Inho was happy. Mina believed Inho was happy.'

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19 In some diachronic studies, there are some Korean linguists who claim that *ta-nun* is not derived from *ta-ko ha-nun*. Lee, P. (1995b) is such a case. It is also known that *ta-nun* is not always replaced with *ta-ko ha-nun*:

i. mina-nun antoy-ess-ta-(??ko ha)-nun {nwuncholi, kisayk, nukkim}-(u)lo
Mina-top pitiful-pst-dec-(cmp do)-adn {eyes, facial.mood, feeling}-with
palapo-ass-ta.
look-pst-dec
'Mina looked with the {eyes, facial mood, feeling} that it was pitiful.'

Here the expression *ko ha* is not acceptable. See Lee, P. (1993, 1995b), Lee, C. (1996), Kim, S. (2004), etc., for more discussions about the two expressions.

Here the expression *ko ha* 'cmp do' is optional.
However, even when a *kes*-phrase with *ta* denotes a fact, *ko ha* can be used:

(45) mina-nun inho-ka haynpokha-ta-(ko ha)-nun {kes, sasil}-ul
Mina-top Inho-nom $_{be}$happy-dec-(cmp do)-adn {thing, fact}-acc
al-ass-ta.
know-pst-dec
'Mina knew (the fact) that Inho was happy.'

A fact does not presuppose an event of assertion. There are other nouns that do not presuppose an event of assertion but can be used with *ta-ko ha-nun*:

(46) oyn son-i yakha-ta-(ko ha)-nun {yakcem, pimil}
left hand-nom $_{be}$weak-dec-cmp do-adn {weakness, secret}
'a {weakness, secret} that his left hand is weak'
(47) nac-i ccalp-ta-(ko ha)-nun uymi
day-nom $_{be}$short-dec-cmp do-adn meaning
'the meaning that the day is short'
(48) mina-ka tolao-l.kesi-la-(ko ha)-nun mitum
Mina-nom come.back-mod-dec-cmp do-adn belief
'the belief that Mina will come back'

This shows that *ta-ko ha-nun* itself does not presuppose that there is an event of assertion. We can say the same thing about *kes*-phrases:

(49) oyn son-i yakha-ta-(ko ha)-nun kes-i ku-uy choytay-uy
left hand-nom $_{be}$weak-dec-cmp do-adn thing-nom he-of highest.degree-of
{yakcem, pimil}-i-ta.
{weakness, secret}-be-dec
'That his left hand is weak is the most serious {weakness, secret}.'

Here there is an event of assertion assumed. Therefore, when a *kes*-phrase is used as the object of *mit* 'believe' with the meaning of assertion, we cannot explain the meaning of assertion by resorting to the use of *ko ha*.[20] Therefore to find a clue, we need to look into the

20 One tentative idea is to assume that the *kes*-phrase denotes a proposition because the adnominal clause is an internally headed relative clause. But this idea is not plausible because the head noun can be replaced

way that the denotation of the adnominal clause is determined.

3.3. Noun-modifying declarative clauses

When a *that*-clause is embedded in an NP in English, Moulton (2009, 2015) analyzes the meaning of a *that*-clause is interpreted as follows:

(50) $[\![$ that Ann is happy $]\!]^{w}$ = ' λx[CONTENT^{w}(x) = λw'[happy$^{w'}$(ann)]]'

Then an NP with a *that*-clause is interpreted as follows:

(51) $[\![$ the rumor that Ann is happy $]\!]^{w}$
= ' ιx[rumor'^{w}(x) $\wedge$ CONTENT^{w}(x) = λw'[happy$^{w'}$(ann)]]'

When there is no overt head noun, he assumes that there is a null functional category which takes a proposition and yields a property of individuals.

This analysis is not applicable to Korean for a few reasons. First, we do not need to assume a null functional category which yields a property of individuals. When a clause needs to occur at the position of an NP, it is overtly embedded in a *kes*-phrase with the adnominalizer *(u)n*. We saw that if a clause is used in the position of the subject of a sentence, it needs to be embedded in an NP:

(52) mina-ka ttena-ss-ta-nun kes-i somwun-uy nayyong-i-ta.
Mina-nom leave-pst-dec-adn thing-nom rumor-of content-be-dec
'The thing that Mina left is the content of the rumor.'

Here the *kes*-phrase does not denote a property of individuals. The word *nayyong* 'content' is overtly mentioned and the *kes*-phrase conveys that content.

More seriously, the distinction of an object and the content is not always clear. One reason for introducing CONTENT is to distinguish an object and a proposition, because a rumour can spread but a proposition cannot. But an NP which is supposed to denote an object may have

with an overt common noun. An internally headed relative clause does not allow a different head noun than *kes*:

i. inho-nun mina-ka tomangka-nun {kes, *salam}-ul cap-ass-ta.
Inho-top Mina-nom run.away-adn {thing, person}-acc catch-pst-dec
'Inho caught Mina, who was running away.'

to denote the content too:

(53) a. yuna-ka yeyppu-ta-nun somwun-un sasil-i ani-ta.
Yuna-nom $_{be}$pretty-dec-adn rumour-top fact-nom not.be-dec
'The rumour that Yuna is pretty is not true.'
b. ??$true^{w}(\iota x[rumour'^{w}(x) \wedge CONTENT^{w}(x) = \lambda w'[pretty^{w'}(yuna)]])$

The truth or factuality is a matter of content, not a matter of (abstract) object like a rumor. If the rumor were necessarily an object, it could not be true or false. This implies that the variable x is always underspecified with respect to whether it ranges over abstract objects or their contents, and further specifications are given by the interactions between the meanings of expressions that are related to the noun.[21] A more plausible analysis should be like this:

(54) $true^{w}(\iota p[rumour'^{w}(p) \wedge p = \lambda w'[pretty^{w'}(yuna)]])$
(understood as $true^{w}(\iota p_{content}[rumour'^{w}(p_{object}) \wedge p_{content} = \lambda w[pretty^{w}(yuna)]]))$

p can range over propositions as denoting abstract objects or their contents, depending on which predicate it is interpreted with, assuming that further specifications are made in a separate component of interpretation that deals with selectional restrictions.

There are also cases where CONTENT is not appropriate:

(55) tam-ul nem-keyss-ta-nun {yongki, nwunchi}
wall-acc go.over-will-dec-adn {courage looks}
'the {courage, looks} that (he) will go over the wall' (Lit.)

(56) tam-ul nem-un huncek
wall-acc go.over-adn trace
'the trace that (he) went over the wall' (Lit.)

In these cases it does not seem that the adnominal clause denotes the content of a courage,

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21 A more typical example is like the following:

i. a. Ann read a magazine while she was waiting. (information)
b. He works for a magazine. (organization)
c. The magazine is thick but interesting. (physical object & information)

When we interpret the word *magazine*, it might be correct to specify whether it denotes the information, or organization or physical object of a magazine. In (i.c), one NP is used with two predicates which require different facets of a magazine.

a look, or a trace. See Yeom (2015, 2017) for more discussions of adnominal clauses in Korean. In order to cover such cases, we need a more general mechanism:

(57) Suppose that α is a clause that denotes a property of situations/facts/propositions and β is a noun that denotes a property of objects. That is,
$[\![\alpha]\!]^w = \lambda s[P(s)]$; $[\![\beta]\!]^w = \lambda x[Q(x)]$. Then
$[\![[_{NP} [_{CP} \alpha] \beta]]\!]^w = \lambda x \exists s[P(s) \wedge Q(x) \wedge R(x,s)]$
(s can be replaced with f or p.)

If an adnominal clause denotes a property of situations, facts, or propositions and a head noun denotes a property of objects, the NP is interpreted as a property of objects which have a contextually given relation R with a situation/fact/proposition.[22]

In many cases, if the clause denotes a proposition, R can be CONTENT, but as I said, we may not need to specify the relation separately. Furthermore, this kind of interpretation does not occur when the head noun is *kes*:

(58) yuna-ka yeyppu-ta-nun {somwun, *?kes}-i tol-ass-ta.
Yuna-nom $_{be}$pretty-dec-adn {rumour, thing}-nom circulate-pst-dec
'The rumour circulated that Yuna is pretty.'

If the head noun is *kes*, the referent of the *kes*-phrase is limited and it cannot denote anything like a rumor. Therefore we can assume that a *kes*-phrase denotes something that the phrase inherently denotes, without the help of a separate relation noun.

We saw that even if *ta-nun* can be paraphrased as *ta-ko ha-nun*, we do not need to depend on the meaning of *ko ha*. We also saw that a declarative sentence with a Mood marker denotes a proposition or a fact. Thus we can propose the following:

(59) a. $[\![[_{MP} \alpha]\text{-nun}]\!]^w = \lambda p[p \models [\![{}^\wedge\alpha]\!]^w]$
b. $[\![[_{MP} \alpha]\text{-nun}]\!]^w = \lambda f[[\![\alpha]\!]^w \sqsubseteq f]$

22 A more typical example is the following:

i. sayngsen-i tha-nun naymsay
fish-nom burn-adn smell
'the smell that a fish burns' (Lit.) ('the smell that arises when a fish burns')

The adnominal clause does not include a structure in which a smell-denoting NP can occur, but the adnominal clause modifies the common noun naymsay 'smell'. See Yeom (2017) for more extensive discussions of how such adnominal clauses are interpreted.

In (59.a), a *kes*-phrase denotes a property of propositions, while in (59.b), the *kes*-phrase denotes a property of facts. Since *kes* is semantically null, the meaning of a *kes*-phrase is the same as the meaning of an adnominal clause. As I said in (33), if a *kes*-phrase is used as an argument, it undergoes a process. When a *kes*-phrase with *ta* denotes a proposition or fact, the plausible process is the ι-operation. That is, there is a proposition or fact that is uniquely determined in the context and that entails the content of the adnominal clause. Which meaning of the two a *kes*-phrase gets also depends on the linguistic context that it occurs.

The meanings of the verbs al_2 and mit_2 can be defined as follows:

(60) a. $[\![\mathrm{al}_2]\!]^w = \lambda f \lambda x \lambda e[\mathrm{know}^w(e,x,f)]$

b. $[\![\mathrm{mit}_2]\!]^w = \lambda p \lambda x \lambda e[\mathrm{believe}^w(e,x,p)]$

These uses of the verbs take a *kes*-phrase, which denotes something that exists independently of the meaning of the verbs. Therefore it can play the role of the object of the verb. And the verbs are eventive because it is presupposed that there is a cognitive contact of the subject with the external object. For this reason, an event variable e is used in the meaning specification.

Now we can interpret the sentences in (25.b) and (26), ignoring the Tenses:

(61) a. $[\![\text{mina-ka ttena-ss-ta}]\!]^w = \exists s[\mathrm{left}^w(s,\mathrm{mina})]$

b. $[\![\text{mina-ka ttena-ss-ta-nun}]\!]^w = [\![\text{mina-ka ttena-ss-ta-nun kes}]\!]^w$

= i) $\lambda p[p \vDash \lambda w' \exists s[\mathrm{left}^{w'}(s,\mathrm{mina})]]$

ii) $\lambda f[\exists s[\mathrm{left}^w(s,\mathrm{mina})] \sqsubseteq f]$

$\Rightarrow$ i) $\iota p[p \vDash \lambda w' \exists s[\mathrm{left}^{w'}(s,\mathrm{mina})]]$

ii) $\iota f[\exists s[\mathrm{left}^w(s,\mathrm{mina})] \sqsubseteq f]$

c. $[\![(25.b)]\!]^w = \exists s[\mathrm{know}^w(s,\mathrm{inho}, \iota f[\exists s'[\mathrm{left}^w(s',\mathrm{mina})] \sqsubseteq f])]$

d. $[\![(26)]\!]^w = \exists s[\mathrm{believe}^w(s,\mathrm{inho}, \iota p[p \vDash \lambda w' \exists s[\mathrm{left}^{w'}(s,\mathrm{mina})]])]$

The adnominal phrases denote a property of propositions or a property of facts, and they become the meanings of the *kes*-phrases. They undergo the ι-operation. Here I assume that the ι-operator does not mean maximality but definiteness, and that definiteness comes from the uniqueness of a familiar antecedent.

Then how is the presupposition of an event of assertion accounted for? If *kes* corresponded to a noun like *cwucang* 'claim/assertion', the presupposition could be easily explained. The meaning of cwucang can be defined as follows:[23]

(62) $[\![\text{cwucang}]\!]^w = \lambda p'\{\exists e \exists x \exists p[\text{assert}(e, x, p)\}[p \vDash p']$

The noun denotes a property of propositions, and it is presupposed that there is an event of assertion, because a proposition arises as a result of the event of assertion.

However, we cannot claim that *kes* can always replace *cwucang* because it is semantically trivial and the restrictions on it are determined by linguistic contexts:

(63) mina-ka ttena-ss-ta-nun {kes, *?cwucang}-i somwun-uy
Mina-nom leave-pst-dec-adn {thing, assertion}-nom rumor-of
nayyong-i-ta.
content-be-dec
'The {thing, *?assertion} that Mina left is the content of the rumor.'

One way to explain the presupposition of an event of assertion might be to claim that a proposition-selecting verb presupposes an event of assertion. A response-stance verb is such a verb.[24] However, the verb *mit* is not a response-stance verb. The reason that an event of assertion is presupposed when a *kes*-phrase is selected by the verb *mit* 'believe' is that a *kes*-phrase denotes a proposition that exists independently of the meaning of the verb *mit*. If someone believes something that exists independently of his or her belief, it is something she heard from someone else or read from something. This requires that there be an implicit or explicit event of assertion. Therefore the subject of the verb *mit*, as well as the speaker, believes that there is an event of assertion.

23 If *cwucang* means a property of events like $\lambda e \exists p \exists x[\text{assertion}(e, x, p)]$, it is not presupposed that there is a proposition as an assertion.

24 There are observations by authors including Cattell (1978), Hegarty (1992), de Cuba (2007), and Kastner (2015) that response-stance verbs embed 'familiar' complements:

i. response-stance verbs: agree, admit, confirm, ...
ii. Response-stance: Embedded clause refers to a familiar idea.
iii. Alice agreed/admits/confirmed [that Ron called]...
#...but no one had said that Ron called.

And de Cuba (2007) and Kastner (2015) observe that in Hungarian and Hebrew, response-stance verbs tend to embed nominal clauses. The same is true of Korean. However, *mit* is not a response-stance verb. Therefore familiarity is not required by the verb.

4. Conclusion

I discussed three verbs of epistemic events or states and their complements. The three verbs can combine with a *ko*-phrase because they all can express an epistemic state. The verb *al* 'know' depends on an external source inherently. For this reason, when it is used with a *ko*-phrase, the verb needs to be interpreted as an epistemic stative verb by attaching an Aspect marker *ko.iss* to the VP.

The verb *sayngkakha* 'think' only expresses an epistemic state and a thought arises internally in the mind of the subject. On the other hand, when a *kes*-phrase denotes a fact or a proposition, it is presupposed that the fact or proposition exists independently of the epistemic state or process expressed by the verb that takes it. Therefore *sayngkakha* cannot be used with a *kes*-phrase. The verb *al* can be used with a *kes*-phrase and this is more natural than with a *ko*-phrase. The verb *mit* 'believe' can take both a *ko*-phrase and a *kes*-phrase, and it can be used to express what the agent hears or whatever belief comes into being in the mind of the subject.

I also claim that *kes* is a noun that is semantically null and a *kes*-phrase expresses what is determined internally. A *kes*-phrase without *ta* denotes a property of situations inherently and can be converted to a fact, if selected as a fact. A *kes*-phrase with *ta* denotes a property of propositions or a property of facts. This is quite natural because a sentence with a declarative Mood marker denotes a (true) proposition. An adnominal ending just plays the role of abstracting over semantic entities, whatever they are.

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